

GCRew_Flume_TidalSurvey_DOC_QAQC_202402

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##Setup - Change things here & write any notes

```
#identify section  
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE  
Date_Run = "02/25/2024" #Date that instrument was run  
Run_by = "Stephanie J. Wilson & Carey Pelc" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = "Standard Curve was assessed manually on the instrument." #any notes from the run  
#samples <- c("Flume") #whatever identifies your samples within the same names  
samples_pattern <- "Flume|50|50 dup|51|52|53|54|55|56|57|58|59|60|60 dup" #paste(samples, collapse =  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_50C_2N" #what did you name your check standards?  
  
library(here)
```

Warning: package 'here' was built under R version 4.3.3

here() starts at S:/Biogeochemistry/People/Steph Wilson/Github/serc-lateral-flux

```
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
raw_file_name = c(here("Tidal_Surveys/Raw_Data/202402", "TOCT_Flume_TidalSurvey_Feb2024_Spring_DOC_1  
  here("Tidal_Surveys/Raw_Data/202402", "2024_0225_DOC_ForStephAvg.txt"))  
  
#file path and name for raw all peaks file  
#raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt" #example  
# raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202411_allpeaks.txt"  
  
#file path and name of processed data file  
#processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202505.csv" #example  
processed_file_name = here("Tidal_Surveys/Processed_Data/202402", "Flume_TidalSurvey_DOC_Processed_1  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
Raw_Metadata = here("Tidal_Surveys/Raw_Data/202402", "Tidal_Survey_202402_Metadata.csv")
```

```
#### This set of samples had some reruns and we need to use them
Sample_IDS <- read.csv(here("Tidal_Surveys/Raw_Data/202402", "202402_DOC_rerun_ids.csv"))

#qac log file path for this year
#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2024.csv"
```

```
##Set Up Code
```

Import Data Functions

Import Sample Data

```
## Import Sample Data
```

```
## New names:
## New names:
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>   <dbl> <chr>
## 1 Flume_20240208_SW-0900    3.60   0.286 2/19/2024 12:23:51 AM
## 2 Flume_20240208_SW-1000    3.62   0.246 2/19/2024 12:49:31 AM
## 3 Flume_20240208_SW-1100    3.08   0.476 2/19/2024 1:14:51 AM
## 4 Flume_20240208_SW1200    3.41   0.495 2/19/2024 1:43:31 AM
## 5 Flume_20240208_SW-1300    3.35   0.528 2/19/2024 2:11:12 AM
## 6 Flume_20240208_SW-1400    3.23   0.543 2/19/2024 2:35:38 AM
```

Assessing Standard Curves - assessed manually on the instrument

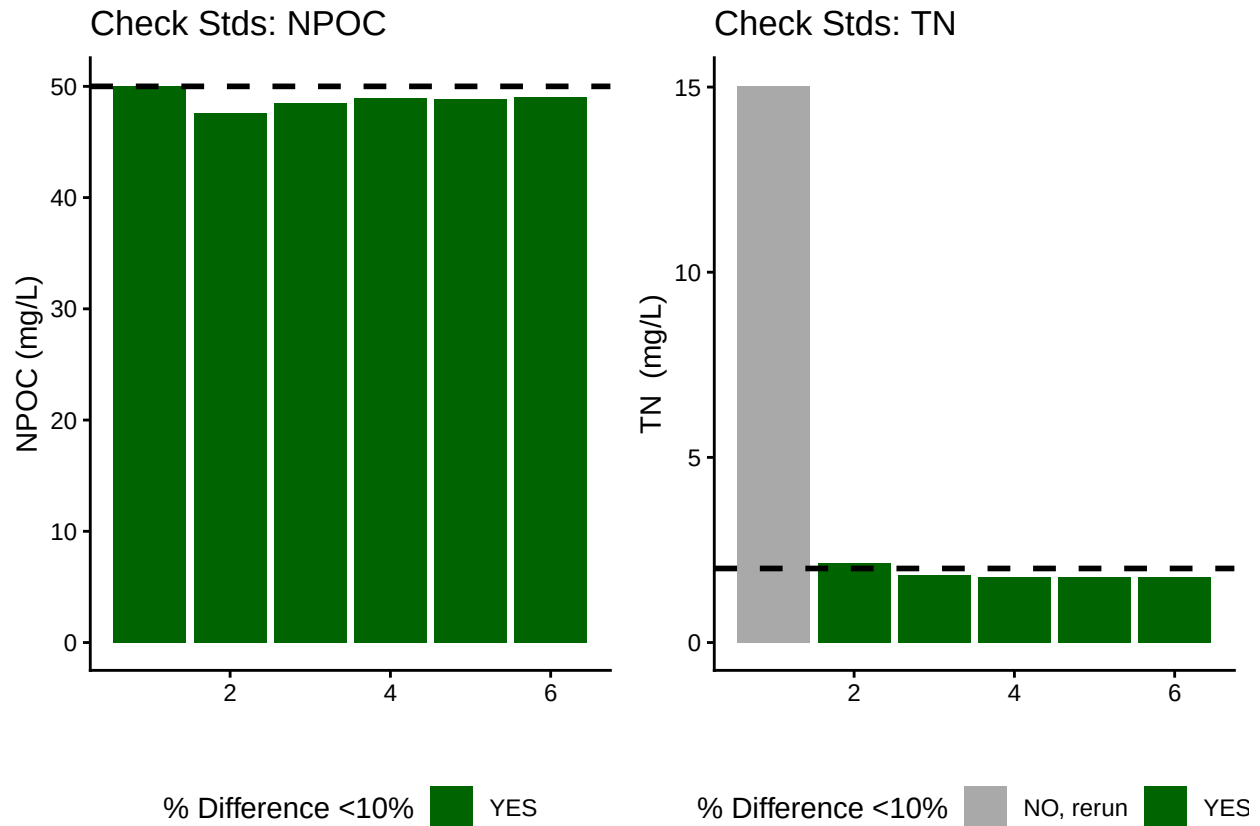
Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

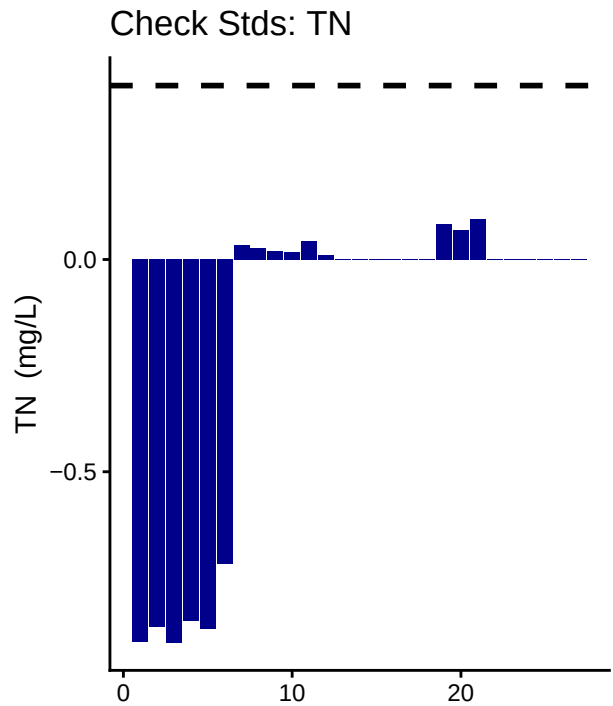
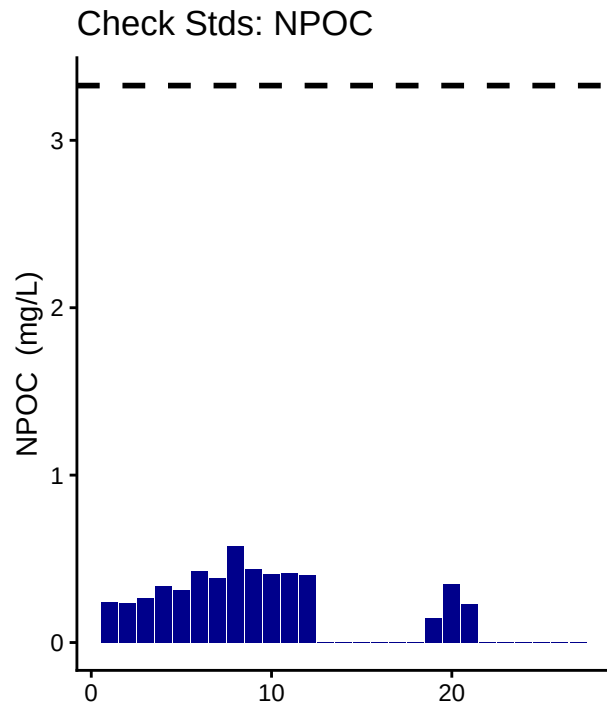
Assess Blanks

```
## Assess Blanks
```

```
## New names:
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ Y

```
## carbon blanks:
```

```
## [1] 0.191837
```

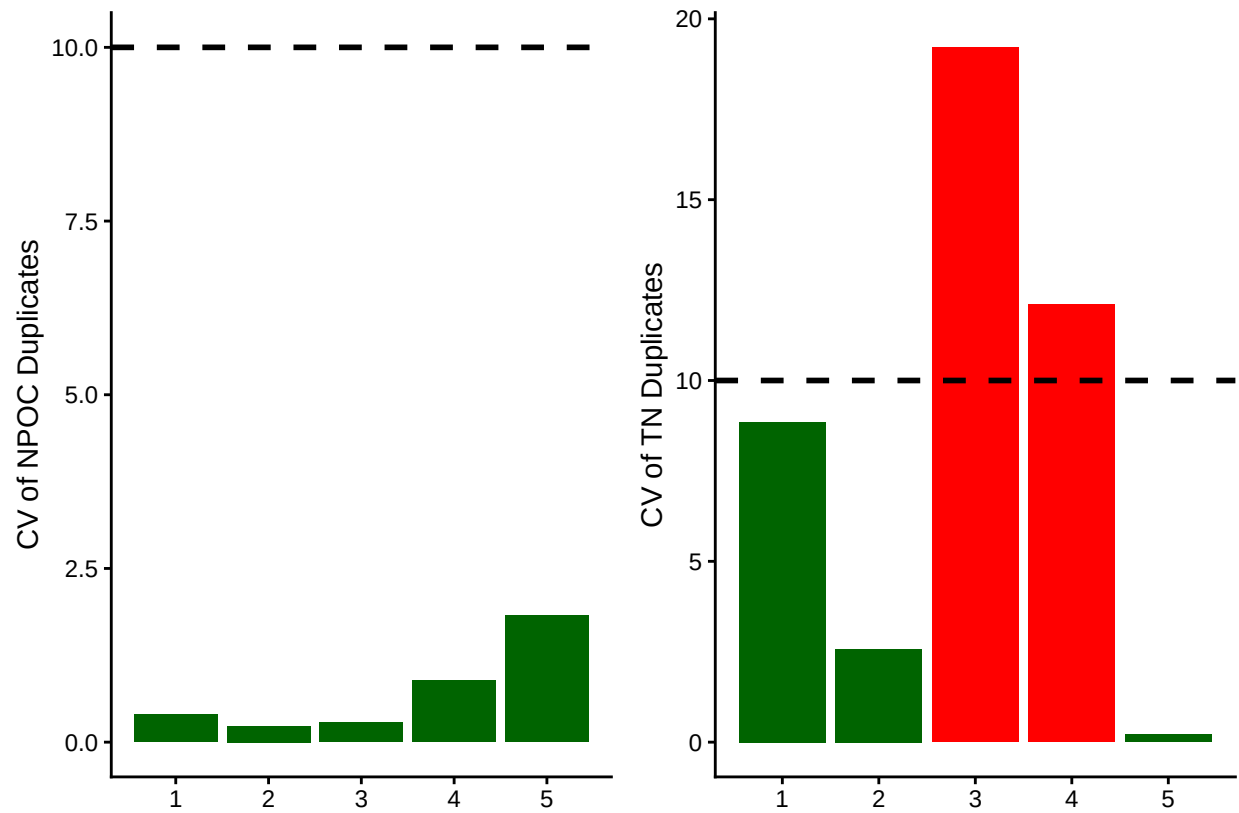
```
## nitrogen blanks:
```

```
## [1] -0.1741278
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

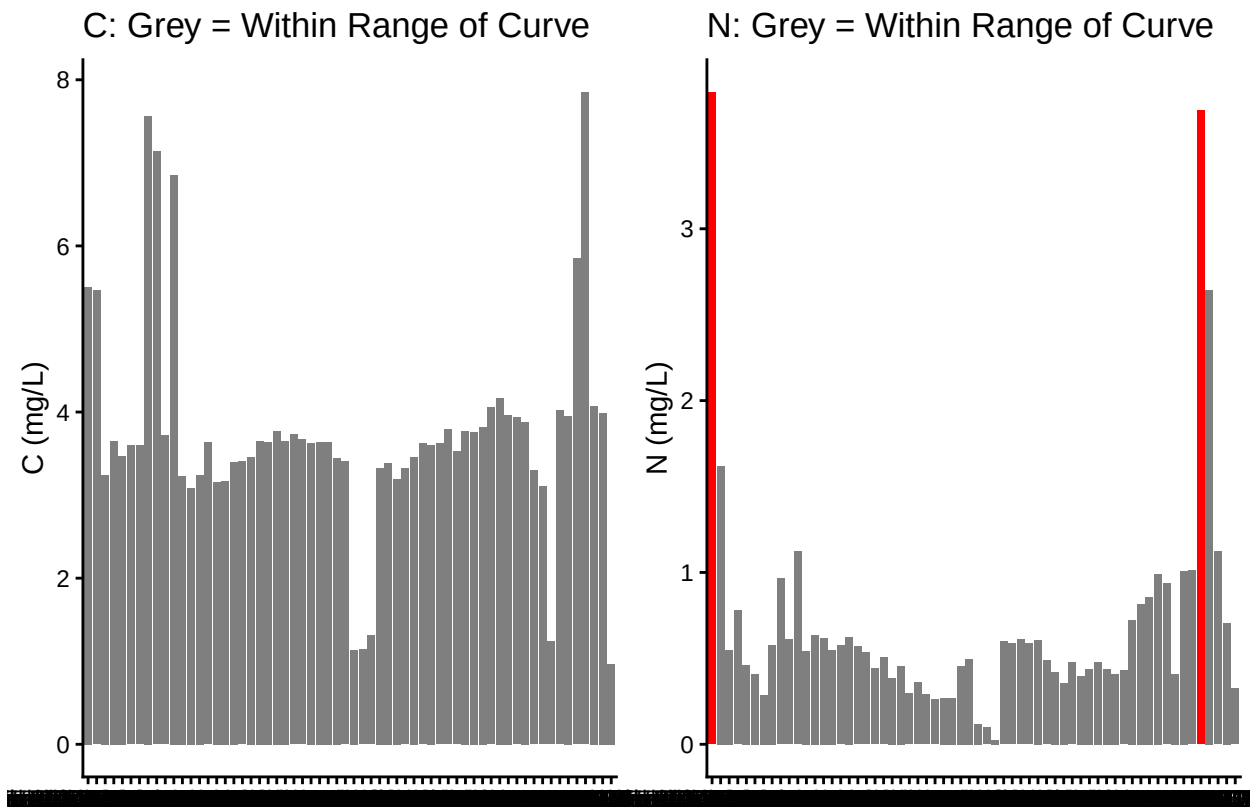


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

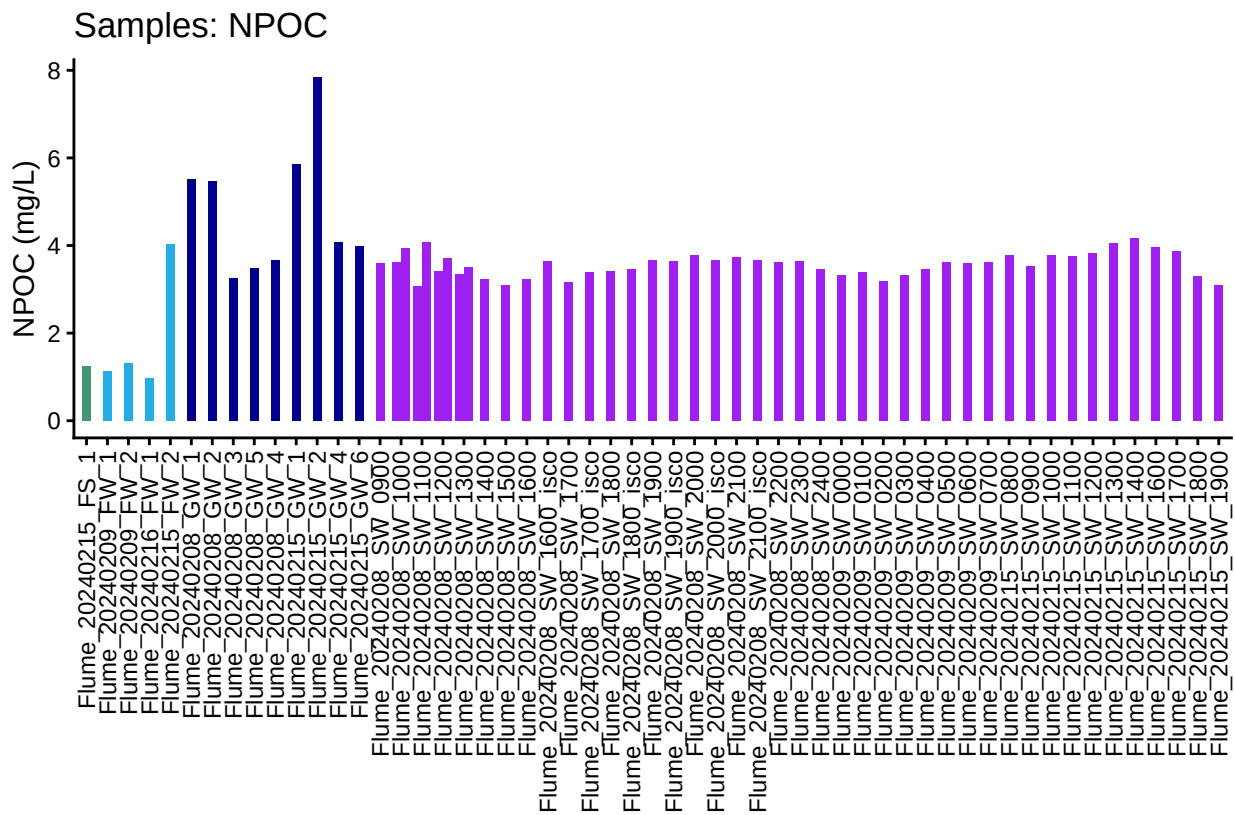
```
## Sample Flagging
```

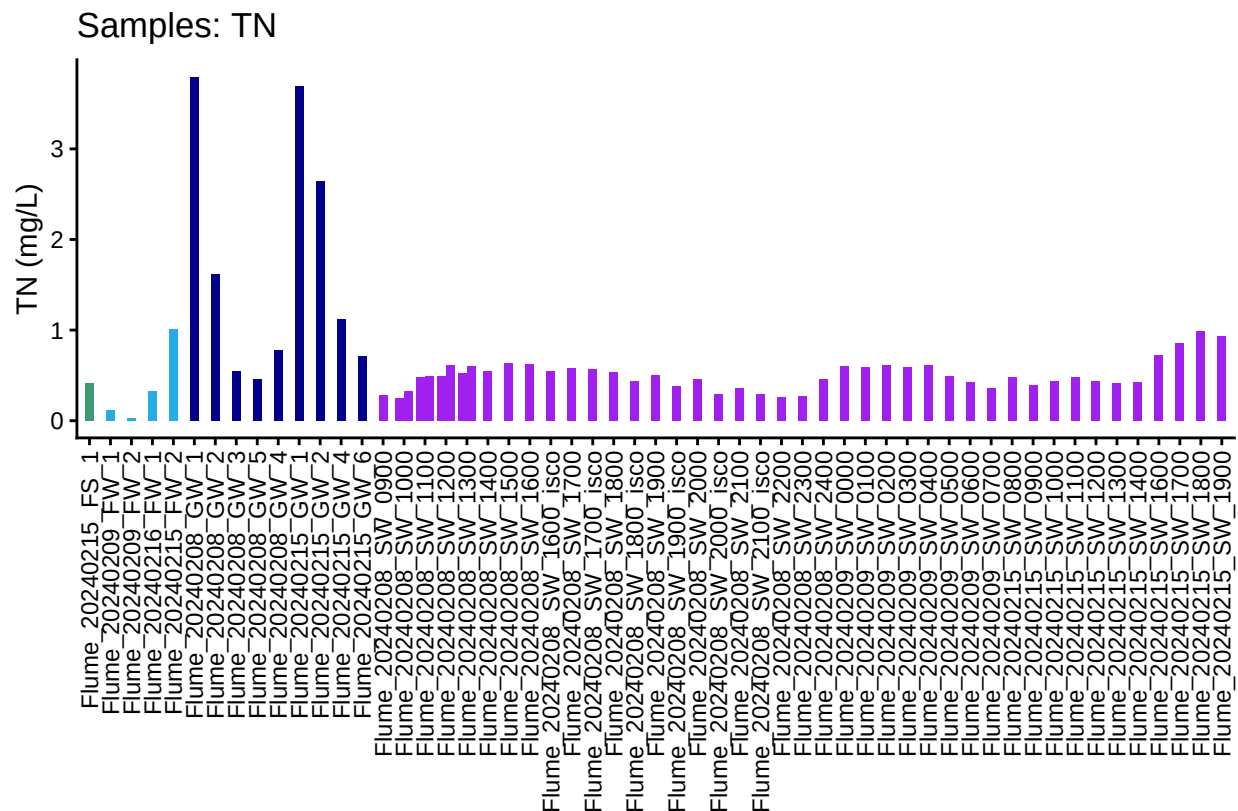


Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("Flume", "20240208", "SW", "0900"), c("Flume", "20240208", :
## number of columns of result is not a multiple of vector length (arg 1)
```





Convert data from mg/L to uMoles/L

Match metadata & merge info

Merge with Metadata

##	Survey_Type	Site	Date	Time_EDT	Time_EST	Sample_Type
## 1	Tidal Survey	Flume	2/8/2024	9:00	10:00	SW
## 2	Tidal Survey	Flume	2/8/2024	10:00	11:00	SW
## 3	Tidal Survey	Flume	2/8/2024	11:00	12:00	SW
## 4	Tidal Survey	Flume	2/8/2024	12:00	13:00	SW
## 5	Tidal Survey	Flume	2/8/2024	13:00	14:00	SW
## 6	Tidal Survey	Flume	2/8/2024	14:00	15:00	SW

##	Sample_ID	DO_mgL	pH	Salinity_ppt	Temp	DIC	DICex	SO4_H2S	NUTR	DOC
## 1	Flume_20240208_SW_0900	<NA>	NA	<NA>	NA	x	x	x	x	x
## 2	Flume_20240208_SW_1000	<NA>	NA	<NA>	NA	x	x	x	x	x
## 3	Flume_20240208_SW_1100	<NA>	NA	<NA>	NA	x	x	x	x	x
## 4	Flume_20240208_SW_1200	<NA>	NA	<NA>	NA	x	x	x	x	x
## 5	Flume_20240208_SW_1300	<NA>	NA	<NA>	NA	x	x	x	x	x
## 6	Flume_20240208_SW_1400	<NA>	NA	<NA>	NA	x	x	x	x	x

##	GHGs	TA	Radium_Vol_L	Notes
## 1	x	x	5	February: Neap Tide
## 2	x	x	5	February: Neap Tide
## 3	x	x	5	February: Neap Tide
## 4	x	x	5	February: Neap Tide


```
## 5      x  x          5 February: Neap Tide
## 6      x  x          5 February: Neap Tide
```

```
## Some sample IDs are missing from metadata.
```

```
## [1] "Flume_20240209_SW_0500" "Flume_20240209_SW_0600" "Flume_20240209_SW_0700"
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 22
##   Project Site Replicate Sample_Type Sample_ID Date Time_EDT Time_EST npoc_mgL
##   <chr>   <chr> <chr>      <chr>      <chr>   <chr> <chr>   <chr>      <dbl>
## 1 SERC L~ Flume 0900      SW      Flume_20~ 2/8/~ 9:00    10:00    3.60
## 2 SERC L~ Flume 1000      SW      Flume_20~ 2/8/~ 10:00   11:00    3.62
## 3 SERC L~ Flume 1100      SW      Flume_20~ 2/8/~ 11:00   12:00    3.08
## 4 SERC L~ Flume 1200      SW      Flume_20~ 2/8/~ 12:00   13:00    3.41
## 5 SERC L~ Flume 1300      SW      Flume_20~ 2/8/~ 13:00   14:00    3.35
## 6 SERC L~ Flume 1400      SW      Flume_20~ 2/8/~ 14:00   15:00    3.23
## # i 13 more variables: npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>,
## #   tdn_uM <dbl>, tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>,
## #   Field_notes <chr>, DO_mgL <chr>, pH <dbl>, Salinity_ppt <chr>, Temp <dbl>,
## #   Radium_Vol_L <dbl>
```

```
#end
```